

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Summer 2021 - 22 Examination

Semester: 5
Subject Code: 03105302
Subject Name: Design and Analysis of Algorithm

Date: 18-04-2022
Time: 10:30am to 1:00pm
Total Marks: 60

Instructions:

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 Objective Type Questions - (Each of one mark) (15)

1. Which of the following Data Structure is linear data structure?
 - A. Trees
 - B. Graphs
 - C. Arrays
 - D. None of above
2. The complexity of bubble sort algorithm is
 - A. $O(n)$
 - B. $O(\log n)$
 - C. $O(n^2)$
 - D. $O(n \log n)$
3. Two main measures for the efficiency of an algorithm are
 - A. Processor and memory
 - B. Complexity and capacity
 - C. Time and Space
 - D. Data and Space
4. What is the type of the algorithm used in solving the 8 Queens problem?
 - A. Greedy
 - B. Dynamic
 - C. Branch and Bound
 - D. Backtracking.
5. Which of the following method is computing total cost of an algorithm in amortized analysis?
 - A. Aggregate method
 - B. accounting method
 - C. potential method
 - D. both (C) and (B)
6. Define Term: Algorithm
7. Define Term: Branch & Bound algorithm
8. Define: NP Hard Problems
9. Define O notation of time complexity
10. Define: DAG

11. The complexity of selection sort algorithm is _____.
12. _____ is an algorithm design technique where to solve a problem; the problem is divided in to sub-problems.
13. Define order of growth
14. What is N-queens problem?
15. What is MST?

Q.2 Answer the following questions. (Attempt any three) (15)

A) Solve the following recurrences using recursion tree method.

$$T(n) = 2T(n/3) + n$$

B) Explain Bellman-Ford Algorithm in reference to Single Source Shortest Path.

C) Explain the Principal of Optimality in Dynamic Programming.

D) All the problems are cover under 'P' class problem and 'NP' class problem. Can we convert a problem of NP class to P class? Justify your answer with suitable example.

Q.3 A) Prove that worst case complexity of quick sort and insertion sort is $O(n^2)$. (07)

B) Discuss 0-1 knapsack problem. Consider an instance of a knapsack problem with four items and weight carrying capacity as $W=5$. Weight and benefit of items are given below. Solve this problem using 0-1 Knapsack. (08)

Weight	2	3	4	5
Benefit	3	4	5	6

OR

B) Compare Naïve String matching and KMP string matching algorithm. (08)

Q.4 A) Mention the differences between DFS and BFS. Write and analyze an algorithm for BFS. (07)

OR

A) Explain divide and conquer approach for Merge sort and analyze it. (07)

B) Find out minimum spanning tree and its length using Kruskal's algorithm for given graph. (08)

